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Ontology for a Knowledge-Based Deconstruction of Buildings Based on BIM Models and Linked Data Principles

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Abstract: The deconstruction of a building marks the end of its life cycle. At the same time, this phase offers many opportunities for the acquisition and reuse of used building materials. During the deconstruction phase, planning and execution are carried out using a variety of different software applications. Despite the use of digital 3D models according to the Building Information Modeling (BIM) method, there is no interoperability along the value chain in the deconstruction of a building. To counteract the lack of interoperability in the deconstruction phase of buildings, the article develops an ontology for the formalization of knowledge within the deconstruction phase and the usability of heterogeneous data sets in connection with the 3D BIM model. After assessing the status quo on the topic of BIM in deconstruction and ontologies in building construction, the systematic development of the deconstruction ontology (DCO) and its validation on a demonstrator is conducted. The result is the presentation of the basic usability of ontologies in conjunction with graphical databases and digital 3D models in the deconstruction phase. This ensures interoperability along the value chain and counteracts data loss.

Keywords: deconstruction; ontology; linked data; buildings; BIM

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1. Introduction

The built environment is a major contributor to global CO_2 emissions [1–3]. In order to achieve the targets set for minimizing greenhouse gas emissions and achieving climate neutrality in the medium term, a rethink is needed in the use and reuse of buildings [4,5]. In this regard, the principles of the circular economy (CE) play an important role in the construction industry for the reuse of materials or components [6–8]. The reuse of building components and construction materials avoids landfill or purely thermal recycling. This means that fewer new building materials and components have to be produced and installed at high material and energy costs [9,10]. For the recycling and reuse of materials in the construction sector according to the principles of the CE, an optimized deconstruction of the building is necessary [11,12]. Many different parameters must be considered. This makes the deconstruction of buildings very individual and complex [13]. In addition to the many different parties involved, the choice of the right deconstruction process also depends on internal and external factors. Internal factors relate to the construction. External factors, in turn, relate to the periphery, such as the environment, weather, etc. At the same time, planning the deconstruction process involves a lot of different data and documents. The choice of deconstruction and planning is a manual process that requires a lot of time and human resources. This can lead to human errors in the process chain. At the

same time, there is a lack of data from previous life cycle phases, which is an additional complication [7,14–19]. This problem can be solved by using digital 3D models of the building based on the BIM method [16,19]. These can also serve as a valid, transparent, and consistent database for the deconstruction phase, considering a single source of truth approach. At the same time, the 3D model contains important data about geometry and semantics, such as dimensions, distances, and material and condition values [20]. In the demolition phase, scientific knowledge is still limited, and practical implementation is restricted. The use of the Industry Foundation Classes (IFC) format in conjunction with the 3D model results in existing limits to the integration, storage, and interpretability of heterogeneous data [21,22]. Furthermore, it is complex to make practical knowledge usable in a formalized style for the planning and implementation of deconstruction measures. Decision-making, such as the choice of demolition method or the analysis of possible reusable materials, is also difficult [16].

To counteract these existing challenges, it is necessary to use ontologies to formalize the knowledge, link it with the 3D model, and create interoperability between different data and systems used in the deconstruction phase and adjacent phases [23–25]. This approach is analyzed in this article, and the Deconstruction Ontology (DCO) is developed, considering existing ontologies and in conjunction with digital 3D models. The new approach will also be tested and validated using a demonstrator. This supports the creation of a continuous, interoperable process chain in the deconstruction phase and adjacent phases. In addition, practical knowledge will be formalized and made usable in digital planning and execution during the deconstruction of buildings.

2. Related Works

2.1. BIM in the Deconstruction Phase

The key function of BIM is to create a comprehensive three-dimensional representation of a data-rich model [26,27]. This model not only provides a visual representation of the project but also embeds critical information that can relate to the physical, functional, and operational characteristics of a building or infrastructure. The data within the BIM model can include materials, dimensions, mechanical systems, structural elements, environmental criteria, and other essential aspects that influence the design, construction, operation, and deconstruction phases [28,29].

Particularly in the deconstruction phase of a building or an infrastructure, BIM can contribute to the provision of information that is crucial for the execution and decision-making of the stakeholders of the deconstruction process. A few examples of the meaning of BIM and information provision can be taken from the following list [28–36]:

- Various project participants: Owners/investors, demolition companies, planners, public authorities, residents/public, utility companies, safety officers, etc., are involved in a demolition project. These parties do not work in isolation but rather in collaboration and require information that can be extracted from a data-based model;
- Detailed Information on Materials: BIM stores comprehensive data about the materials used in a specific building, including types, quantities, and locations. Such information is needed, for example, to assess the level of contamination by type and quantity in a deconstruction project;
- Planning and Sequencing: BIM supports creating detailed deconstruction plans, including the sequence for the deconstruction of a building. This ensures that disruptions are reduced by means of forward-looking visualization and that components are removed in a logical and contamination-safe manner;
- Cost Management: Demolition and disposal of materials have a major impact on costs. Costs are influenced by the type and quantity of material as well as the type of

demolition. BIM helps to estimate costs by providing the aforementioned information. Stakeholders can use these data to choose a cost-efficient way of handling the components of a building;

- Waste Management: The classification of material, especially with regard to potential hazards, can be determined in advance based on a digital, information-loaded representation of a building. This information can include the location of hazardous materials and provide further information on the load-bearing capacity of building components that need to be handled carefully during the deconstruction process [28–36];
- Documentation and Compliance: Particularly with regard to relevant German regulations in the focused knowledge domain of the deconstruction, it is necessary to fulfill the requirements and obligations of these regulations. These obligations are associated, for example, with sufficient documentation and verification of the type and whereabouts of waste, which can be achieved using a model.

As the above points reveal, a large amount of information is required for the planning and execution of a deconstruction project. BIM can help to create a basis for the structured, component-related storage of information and, thus, the framework for making decisions [34,35]. BIM approaches in the deconstruction phase have been researched by a number of authors in recent years, resulting in an increasing establishment in the life cycle with a transfer into practice.

In the publication by Aziminezhad and Taherkhani, the current status of the use of BIM in the deconstruction of buildings is elaborated using a literature analysis and shows a wide range of possible applications [34]. Specifically, Akbarnezhad et al. use the possibility of the BIM method for a sustainability assessment of various deconstruction strategies in the preliminary stages of planning and implementation [37]. In addition, Vullings et al. define important guidelines for a BIM-supported audit prior to deconstruction [38]. Since not every building is suitable for a simple deconstruction, Akinade et al. develop a procedure for the preliminary evaluation of the basic deconstruct ability of the building, using various deconstruction methods, external and peripheral parameters, and a digital 3D model as a database [33].

Subsequently, the BIM method is also used in the planning of demolition projects. Thus, Kim and Kim show an application system for decision support in the planning of demolition measures based on digital 3D models [39]. Marzouk and Elmaraghy describe the extension of planning with lean management approaches [40]. Lima et al. published an approach focusing on the BIM-based planning of a demolition project, considering the recyclability and reuse of materials [41]. Various approaches also exist in the transition from the planning to the execution phase. In another work, Marzouk et al. show the possibility of defining lean-orientated processes for the execution of deconstruction tasks [42]. Akinade proposes a framework and software tool for the use of BIM in the execution of deconstruction measures [28]. Van den Berg et al. use a similar approach for BIM-based deconstruction execution, but the authors focus on finding optimal methods for selective deconstruction of complete building components without a major impact on the environment [35]. Sanchez et al. also explore the deconstruction of whole building components and their reuse using IFC format as an open data exchange format to create interoperability in the deconstruction process, with a precise determination of the required parameters [29].

A specific and important scenario in the deconstruction phase is also the handling of the resulting waste. In their work, Han et al. defined a research agenda for the use of BIM in the deconstruction phase, taking waste management into account based on previously published findings [32]. In the same subject area, Nikmehr et al. present the relevant foundations for the development of a special BIM-based software tool for waste management in the deconstruction phase of building structures based on an investigation [43].

2.2. Ontologies for Buildings and the CE

In recent years, the interest in topics relating to linked data and the semantic web in the construction industry has increased. A generic extract of existing ontologies for building construction and then specifically for the area of the CE is presented below. There are ontologies at a meta-level to describe entire domains or to represent a very specific sub-area in building construction along the life cycle of buildings [44–46]. El-Diraby and Osman focus on the basic framework conditions and requirements for structures and contexts in the construction industry [47]. Furthermore, with the e-COGNOS ontology, Lima et al. present a representation and utilization possibilities of knowledge for construction sites [48]. At the same time, domain-specific ontologies exist specifically for the design [49,50], execution [51–54], and operation [55–57] of buildings. In addition, Zheng et al. supplement the previous work on knowledge representation in construction with an extension to include aspects of digital construction. This is represented in the Digital Construction Ontology (DiCon) [58]. With the steady establishment of the BIM method and the use of digital 3D models in the life cycle of buildings, the importance of interoperability in the processes and between systems has also increased [59]. The Industry Foundation Classes Ontology Web Language (IFCOWL) exists for this purpose, considering the Linked Data Principles. This includes the option of representing the IFC structure of a building in the Resource Description Framework (RDF) schema [60,61]. As the complexity of the IFC structure also leads to limitations in comprehensive use in the context of IFCOWL, the Building Topology Ontology (BOT) was developed as an alternative to the standardized and simplified description of building topologies [21].

In addition to the reference to building construction and buildings, the DCO also includes content in the context of sustainability and the CE. In this area, Kebede et al. present a modular ontology for the development of material passports. This is used for effective information management of sustainability data on construction products along the life cycle and identification of potentials of the CE for these [62]. A similar approach is also presented by Soman et al. They extend the view of the reuse of individual building components for new construction project sites to reuse in the planning of entire cities [63]. Sauter et al. developed a further approach to the material passport segment. The authors developed two interdependent ontologies in the context of building material passports to evaluate their reuse. The ontologies annotate materials present in the building according to a CE classification scheme and describe the elements required for material exchange, such as actors, activities, and the reference points of these activities (resources, products, tools, and waste) [64]. In addition to the focus on conventional, in situ construction sites on-site, the article by Kedir et al. looks at the potential of ontology-based material passports for strengthening the CE in modular construction, focusing on industrially manufactured building systems and modules along their life cycle [65]. This ranges from planning to production in the factory, assembly, disassembly, and reuse. Akbarieh et al. are developing an approach for a material database for the deconstruction and reuse of components [66]. The examples described above show that a variety of ontologies exist for the representation of buildings, the process of planning, construction, and operation, or the reuse of building materials.

As mentioned above, there are already isolated ontologies in the area of material reuse, the use of material passports, and the linking to building components in the life cycle [64,66]. Process-related aspects of the planning and implementation of dismantling measures, the required mechanical and personnel resources, and legal and occupational safety parameters have not yet been considered. This research gap is closed in this article with the development of the DCO. At the same time, the DCO is linked to other existing ontologies in the context of CE. The development and validation of the DCO focuses pri-

marily on the creation of interoperability and the use of the RDF data schema, considering the linked data approaches.

3. Methodology

In the previous chapters, the related work in the areas of BIM in the deconstruction phase and ontologies in building construction was identified by conducting a literature analysis, and the status quo was defined. At the same time, the research problem was derived from the findings, and the objective of the article was specified. The following part focuses on the development and validation of the ontology. The process of ontology development follows the Linked Open Terms (LOT) methodology of Poveda-Villalón [67]. The approach is an established way to develop specific ontologies in engineering and, at the same time, make them immediately usable for direct application in practical application scenarios [67]. Furthermore, it is divided into four steps—(i) specification of the ontology requirements, (ii) implementation of the ontology, (iii) publication of the ontology, and (iv) validation (see Figure 1). The LOT methodology is considered established in the development of ontologies, easy to understand, and particularly suitable for practice-oriented ontologies. At the same time, it considers the entire process chain from requirements definition, conceptualization, implementation, and subsequent maintenance and focuses on the reuse of existing ontologies [67]. In addition, it is already being used in other works in the field of ontology development in the construction industry [45,68].

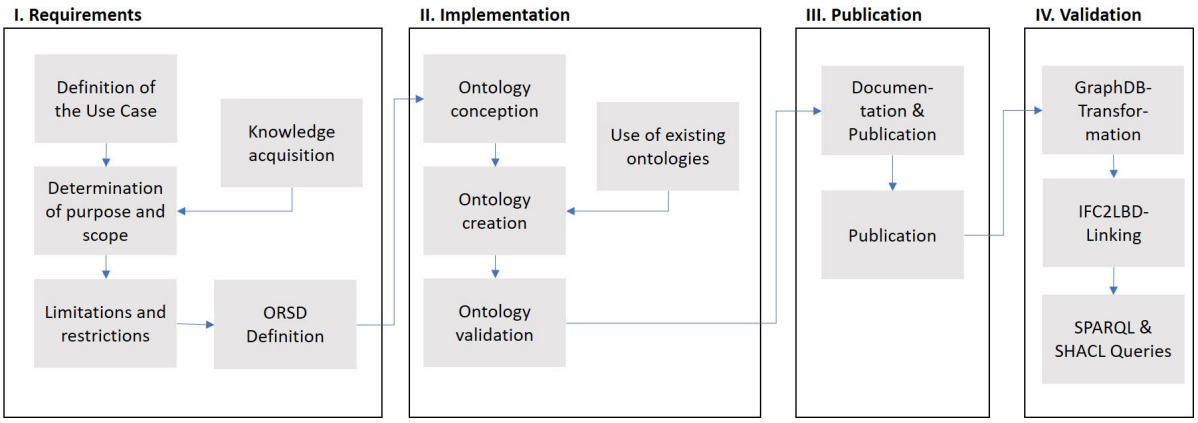


Figure 1. Methodology of the DCO development (based on [67]).

In the first part, the specification of the ontology requirements, the use case of the ontology is defined, the purpose of the ontology is determined, and relevant data sources are identified. This is followed by the conceptualization, implementation, and publication of the ontology. The final step is the validation to test its usability. The validation scenario contains the transformation of the SODIA ontology into a graphical database and linking it to a BIM model of a bridge. The BIM model is first transformed from the IFC data schema into the RDF data schema using a converter [69]. For validation in this scenario, four queries are performed with SPARQL Protocol and RDF Query Language (SPARQL) three constraints using the Shapes Constraint Language (SHACL). The ontology is published and available for open access via the link http://purl.org/dco-ontology [70].

4. Development—DCO

4.1. Requirement Specification

The first step in the development of DCO is the definition of requirements. This includes the description of the application scenario analyzed and the specification of the primary purpose. Furthermore, functional and non-functional requirements are defined.

The DCO focuses on the deconstruction of buildings. This addresses the planning and implementation of deconstruction activities, the necessary regulations to be observed, and the qualifications of personnel. Aspects of the CE are also analyzed, including the reuse of materials or entire components. In addition, dependencies between individual deconstruction processes and the degree of reutilization possibilities are integrated. In addition, the interface between the operational phase of a building, the deconstruction phase, and the planning of a new building is analyzed. The ontology is designed for all stakeholders involved in the demolition phase and adjacent life cycle phases (operating phase and initiation phase of a new project). This includes the building owner, architectural and engineering firms, deconstruction companies, building authorities, and waste disposal companies. The development of DCO is based on a top-down approach. This approach represents the definition of the ontology on the basis of an explicit knowledge base in the form of scientific articles, standards, and technical regulations [71]. No subjective interpretations by the developer are used. The ontology is based on these German standards and technical regulations in the field of deconstruction and the CE. It is based on the published standards and the regulations (e.g., [14,72–75]).

In the next sub-step, the functionalities of the ontology are specified. Both functional and non-functional parameters are defined. The functional requirements are derived from the purpose and the use case under consideration. These are used to develop competency questions for the ontology, which should be answered reliably during the subsequent validation. The following competency questions are defined for the DCO development:

- Which area of application is addressed by the individual deconstruction methods?
- Which deconstruction methods have the lowest impact on the environment and sustainability?
- What equipment is required for an appropriate deconstruction method?
- What safety measures are required for the deconstruction process?
- Which damages have an influence on the structural safety of the building?
- Which components can be reused in terms of the CE, and which components are particularly harmful?
- Which building components are load-bearing and non-load-bearing for the guarantee of structural safety during deconstruction?

The non-functional requirements are based on previously published studies from El Gohary et al. [76], Zhou et al. [77], Costin and Eastman [78], and Seiß et al. [45]. The seven non-functional requirements—(i) coverage, (ii) consistency, (iii) clarity, (iv) conciseness, (v) usability, (vi) extensibility, and (vii) reusability—are used to define the competencies. In addition to objective testing on the demonstrator, these are used for further subjective validation of its functionalities and usability. DCO utilizes existing ontologies in the field of building construction and the CE and expands the missing components in connection with the existing processes in the deconstruction phase. This includes the planning of deconstruction measures with the selection of suitable deconstruction methods and the factors to be considered. Hazardous substances and other safety aspects are also integrated. On top of this, the required qualifications of the personnel and equipment for implementation are considered.

4.2. Use of Existing Ontologies

The development of ontologies requires the use of existing ontologies. That implies the usage of already formalized knowledge of specific subject areas [79–81]. These are identified in the next step of the development process and then linked to the DCO and to each other as part of the subsequent development process. This allows dependencies between the deconstruction phase and other parameters from different domains to be visualized. For

example, this could be a dependency of components on materials and their circularity or existing hazardous substances in building materials and the appropriate deconstruction process. A total of seven ontologies are used in the DCO development process (see Table 1). These integrated ontologies are adapted as required and new links between the individual classes are defined in order to avoid contradictions and redundancies and thus ensure practicable usability.

Table 1. List of integrated ontologies for the DCO.

Position	Name	Prefix	Source
1.	Building Element Ontology	beo	[82]
2.	Building Topology Ontology	bot	[83]
3.	Building Product Ontology	bpo	[84]
4.	Digital Construction Ontologies—Building Materials	dicbm	[85]
5.	Decommissioning and Reuse Ontology	dor	[86]
6.	Damage Topology Ontology	dot	[87]
7.	Friend of A Friend Ontology	foaf	[88]

The Building Topology Ontology (BOT) defines the relationships between the subcomponents of a building [21]. Complementing this, the Building Element Ontology (BEO) classifies building elements such as walls and ceilings within a hierarchical structure and integrates spatial concepts for a more detailed description of these elements [82]. The Building Product Ontology (BPO) provides a reference model for the structured and schematic representation and categorization of construction products in the Architecture, Engineering, and Construction (AEC) domain [84]. The Digital Construction Ontology for Construction Materials (DICBM) is used to describe material properties and their interrelations within construction projects [85]. Focusing on sustainable construction processes, the Decommissioning and Reuse Ontology (DOR) models strategies for the deconstruction, reuse, and recycling of construction materials [86]. The Damage Topology Ontology (DOT) enables the semantic representation and analysis of damage to building components, including the type and spatial localization of damage [89]. Lastly, the Friend of a Friend Ontology (FOAF) semantically links information about people and their relationships across digital platforms [88]. The detailed integration of these ontologies into the DCO ontology is described in Section 4.3.

Further ontology elements have been added through the use of explicit knowledge from standards, technical regulations, and leaflets from the domain of deconstruction planning and execution (see Table 2).

Table 2. Sources for explicit knowledge in the domain—deconstruction.

Position	Name of Article (Translated from German to English)	Source
1.	DIN 18007:2022-09 Demolition works—Concepts, procedures, fields of application	[75]
2.	DIN 4150 Vibrations in buildings—Part 2: Effects on persons in buildings	[90]
3.	DIN 4150 Vibrations in buildings—Part 3: Effects on structures	[91]
4.	DIN 4102 Fire behavior of building materials and building components—Part 1: Building materials; concepts, requirements, and tests	[92]
5.	VDI 6210 VDI 6210—Sheet 1: Demolition of structural and technical installations	[93]
6.	Demolition and dismantling sector (DGUV Rule 101-603)	[94]
7.	Technical Rules for Hazardous Substances (TRGS) 500: Protective measures	[95]

Table 2. *Cont.*

Position	Name of Article (Translated from German to English)	Source
8.	Technical Rules for Hazardous Substances (TRGS) 519: Asbestos, demolition, renovation, or maintenance work)	[96]
9.	Report: Hazardous substance asbestos	[97]
10.	Report: Data analysis on the treatment of mineral construction and demolition waste in Austria—Detailed study on the Federal Waste Management Plan)	[98]
11.	Report: Assessment of Buildings and Constructions (ABC)—Disposal Measures for the disposal properties of buildings and constructions for life cycle assessment	[99]
12.	Report: Construction guidelines for recycling: Working aids for handling construction and demolition waste and for the use of recycled construction materials on federal properties	[74]
13.	Dismantling, recycling, and recovery of ETICS: Possibilities for reusing components of ETICS after dismantling by feeding them into the production cycle of insulation materials or downcycling them into the production of low-value goods through to energy recovery)	[14]
14.	Demolition of residential and office buildings: guidelines	[100]
15.	Controlled demolition for land recycling: resource-conserving demolition of buildings and industrial plants	[73]

4.3. General Structure and Ontology Development

4.3.1. General Structure

Selective deconstruction is characterized by the careful removal of pollutant-containing materials and the systematic separation of construction materials by type [101]. To carry out these measures efficiently, precise knowledge of the location and composition of materials within the building is essential. This includes identifying materials based on floors, rooms, and building components, which aids in the planning of logistical processes, such as the safe transportation of hazardous substances. Additionally, the surrounding environment must be considered to assess potential emissions and their impacts, which is crucial for planning safety measures during deconstruction [102].

In this context, the DCO ontology is developed. It integrates various domains of knowledge to systematically plan the deconstruction of buildings and supports the use of materials in line with the CE principles. The DCO combines information on the spatial structure of buildings through the integration of building topology ontologies. Simultaneously, material properties are captured, including pollutants and other hazardous substances that must be properly handled and disposed of during deconstruction. Building damage is also considered, as it can influence deconstruction methods and necessitate additional safety measures. A key component of the DCO is its coverage of various deconstruction methods. It provides a structured classification of methods that can be selected based on the specific requirements of the project. These include mechanical, thermal, hydrodynamic, and other deconstruction techniques, which are differentiated by criteria such as emission behavior, material type, and area of application. The DCO aids in selecting the most appropriate method by linking it to specific building materials and components. Additionally, it helps to identify sustainable disposal and recycling pathways, maximizing the potential for material reuse.

The DCO ontology implements this comprehensive, deconstruction-specific domain by integrating various ontologies and supplementing them with its own concepts. Figure 2 shows an excerpt from the DCO ontology, where different domains are interconnected through the DCO.

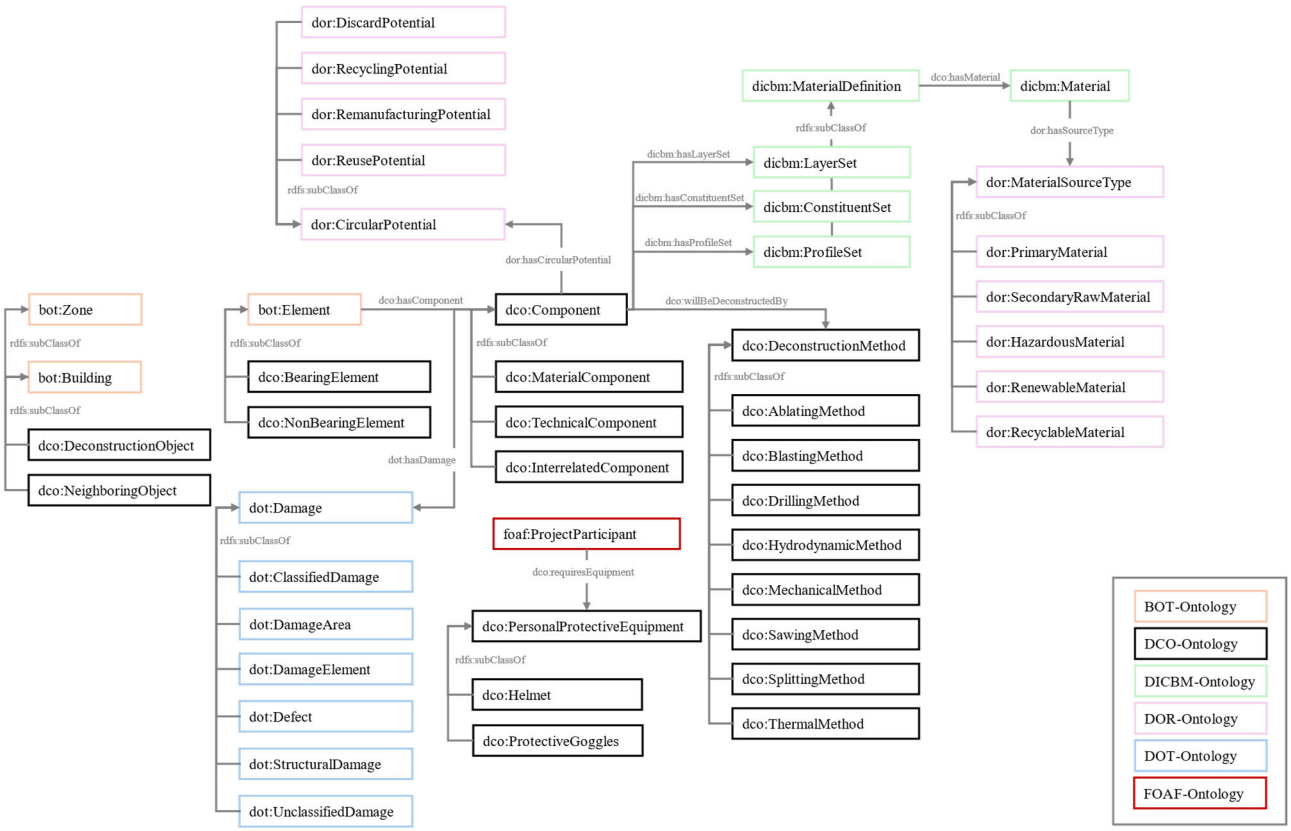


Figure 2. General Structure of the DCO.

4.3.2. Building Topology and Surroundings

For the semantic representation of building topology, the BOT ontology is integrated, which defines the relationships between a building's sub-elements. The BOT ontology provides a detailed description of building topology, including floors, rooms, and the building elements contained therein. It initially distinguishes between different zone types, such as buildings, floors, and rooms. Object relationships like *bot:adjacentZone* enable the semantic linking of adjacent zones within a building [21]. A modification of the *bot:Building* class, a subclass of *bot:Zone*, is necessary because multiple building types are considered in a deconstruction project. While the BOT ontology does not include further subclasses of this type, the DCO ontology includes subclasses for Deconstruction Object (*dco:DeconstructionObject*) and Neighboring Object (*dco:NeighboringObject*). These subclasses are defined as disjoint, meaning a deconstruction object cannot simultaneously be a neighboring object. Moreover, the object properties *dco:directBuildingAdjacency* and *dco:indirectBuildingAdjacency* are introduced to provide a semantic link between the two classes. Both properties have the domain *dco:DeconstructionObject* and the range *dco:NeighboringObject*. The property *dco:directBuildingAdjacency* associates a deconstruction object with one or more neighboring buildings that are directly adjacent, i.e., without any intervening airspace. In contrast, the property *dco:indirectBuildingAdjacency* associates a deconstruction object with neighboring buildings that are indirectly adjacent, meaning that they are within the sphere of influence of the deconstruction object without direct physical contact.

Additional properties, represented in the DCO ontology by data properties, are required to describe the deconstruction object as follows:

- *dco:constructionPeriod*;
- *dco:constructionMethod*;
- *dco:constructionType*;
- *dco:riskOfCollapse*;

- *dco:typeOfUse*;
- *dco:Safeguard*.

In addition to zoning, the BOT ontology includes the *bot:Element* class. Building components can represent both essential parts of a building and individual components of those parts. In practice, distinguishing a building component based on its load-bearing capacity is considered relevant for deconstruction [103]. This distinction is integrated into the DCO ontology by incorporating the classes *dco:BearingElement* and *dco:NonBearingElement* as subclasses of *bot:Element*. Since an element cannot simultaneously be load-bearing and non-load-bearing, these classes are defined as disjoint. Finally, the BOT ontology includes the *bot:Interface* class, which is relevant because it describes a relationship between two building components and cannot be unambiguously assigned to a single component [21].

To structure and functionally differentiate building components, the *dco:component* class is created with the following three subclasses: Material Components (*dco:MaterialComponent*) describe the layered and material-specific structure of a building component, such as masonry and plaster of an exterior wall. The second subclass includes components of technical building equipment (*dco:TechnicalComponent*), such as pipes and wiring contained within a building component. The third subclass, Interrelated Components (*dco:InterrelatedComponent*), consists of multiple materials but is evaluated as a whole. This structure extends the BOT ontology with a deconstruction-specific differentiation of building components and supplements the existing *bot:Zone*, *bot:Element*, and *bot:Interface* classes. To assign a component to a building element, the object property *dco:hasBuildingComponent* is developed. This property encompasses three sub-processes. The first, *dco:exteriorComponent*, associates a building element with a component oriented toward the external environment. The second, *dco:interiorComponent*, links a component positioned within the enclosed space. Lastly, *dco:intermediateComponent* defines a component situated between two other elements in the building structure.

In addition to the BOT ontology, the BEO ontology is integrated, which is based on the IfcBuildingElement subtree of the IFC specification and provides a uniform structure for building components, such as walls and ceilings, organized in a common class hierarchy. It describes the properties of these elements and integrates concepts from the BOT ontology to represent areas, spaces, properties, buildings, and floors, enabling a more detailed description of building components within the framework of a deconstruction project [82].

4.3.3. Damage Types and Consequences

Selective deconstruction is characterized by a variety of activities performed by humans and machines within a building [104]. Damage can become relevant in different situations. Damage such as deformations or fire impacts can compromise the load-bearing capacity of building components and introduce additional hazards due to the forces generated by individual deconstruction methods. Cracks may lead to the release of material-embedded pollutants from the enclosed environment. Moreover, cracks and external influences such as moisture can indicate new damage, such as corrosion. Moisture also promotes the formation and growth of pollutants like mold [105] or dry rot [106]. An ontology for describing damage is the DOT ontology, developed by Al-Hakam Hamdan and Matthias Bonduel, which defines damage in its various impacts and areas and enables the semantic representation of a damaged building component [89].

The primary goal of integrating damage into the DCO ontology is not to provide a detailed damage report or to initiate necessary repair measures. Instead, the focus is on assessing the impact of damage on structural stability and the risk of pollutant release or intrusion. The aim is to represent the hazard and, if necessary, initiate temporary measures

to protect personnel involved in the deconstruction project. Nevertheless, the entire DOT ontology is integrated to cover all possible damage scenarios.

4.3.4. Building Material Classification

To represent the material properties of building components, the DICBM ontology is integrated into the DCO ontology as part of the Digital Construction Ontologies. The goal of these ontologies is to capture relevant objects and properties that actors or systems refer to in the management and execution of construction and renovation projects. This includes physical and spatial entities, actors, or activities [85]. The DICBM ontology describes the essential relationships between building components, construction details, materials, and their properties. The core of the ontology consists of the classes *dicbm:Material*, *dicbm:MaterialDefinition*, and *dicbm:MaterialType*. The integration of the DICBM ontology offers the advantage of synergy with the BOT ontology, which has been expanded to include parts of the DCO ontology. For example, an instance of the *dco:Component* class can possess a *dicbm:LayerSet* (a subclass of *dicbm:MaterialDefinition*) through the object property *dicbm:hasLayerSet*, meaning an element consists of multiple material layers, each with different thicknesses [85].

Accurate classification of building materials is essential for deconstruction projects. By breaking down these materials through the DICBM ontology, building components can be described in detail with their material properties. In the case of utilizing deconstruction ontology in a graph database, the DICBM ontology is partially modified. The class *dicbm:MaterialProperty* and its respective sub-object properties of *dicbm:hasProperty* have been conceptually adapted and implemented as hierarchically structured data properties in the DCO ontology. For example, the data property *dco:materialChemicalProperty* now has sub-data properties such as *dco:hasPHLevel* or *dco:hasAlkalinityConcentration*. This adaption allows the Linked Data Modeler to easily attach additional material information to converted IFC objects via data properties without delving into deep class modeling. Additionally, synthetic instances for individual material properties do not need to be created to link them with the converted IFC objects in the graph database. The range of data properties is primarily modeled using the data types *xsd:Literal* or *xsd:Boolean*. Furthermore, additional data properties for deconstruction-relevant material properties are developed in the DCO ontology. Here, the evaluation of material-embedded pollutants is used as a property and represented by the data property *dco:contaminatedProperty*. Due to the large variety of materials and the differentiation practiced in the field, the data property is divided into five sub-data properties based on chemical components or compounds [107] as follows:

- Non-halogenated hydrocarbon (*dco:hasNonHalogenatedHydrocarbon*);
- Halogenated hydrocarbon (*dco:hasHalogenatedHydrocarbon*);
- Heavy metal (*dco:hasHeavyMetal*);
- Asbestos (*dco:hasAsbestos*);
- Artificial mineral fiber (*dco:hasArtificialMineralFiber*).

4.3.5. Avoidance, Disposal, and Recycling Pathways—Circular Potential

The representation of avoidance, disposal, and recycling methods aims to outline possible pathways for handling deconstructed materials. The DOR ontology is integrated for this purpose. This ontology also addresses the deconstruction process and additionally the reuse of materials. Unlike the DCO ontology, it does not focus directly on specific deconstruction methods but rather on sustainable deconstruction and reuse strategies that enable material reuse. It seeks to integrate the various perspectives of stakeholders and provide a flexible structure for exchanging relevant information on circular poten-

tial [66]. The classes *dor:MaterialSourceType*, *dor:ConnectionType*, *dor:CircularPotential*, and *dor:CircularRequirement* are of particular relevance to the DCO ontology. These classes can supplement the previously introduced material domain with additional information. For instance, the class *dicbm:Material* can be linked to *dor:MaterialSourceType* via the object property *dor:hasSourceType*, indicating whether the material originates from a primary or recycled source. Similarly, it can indicate whether the material is chemical or mechanical. Circular potential can then determine whether the deconstructed object is recyclable and whether it meets circular requirements, such as certifications or specific logistical needs.

4.3.6. Deconstruction Methods

In the context of deconstruction, various methods exist for the progressive separation and deconstruction of building components into individual parts [74,102]. The following class hierarchy is, therefore, designed to provide a structured representation of deconstruction methods and serves as a reference point for assigning properties to these methods. Simultaneously, these classes act as a reference for associating specific deconstruction methods with individual building components.

To distinguish between deconstruction methods, various criteria, such as emission behavior and application areas, can be employed in accordance with the top-down approach. Additionally, the methods can be categorized based on the type of force transmission or the mode of demolition. Given the multitude of deconstruction methods [74,102], the primary objective is to reduce the complexity of the class hierarchy.

In this context, and in alignment with DIN 18007, a combined approach based on the form of force transmission and the type of demolition is used to construct the class hierarchy. Although this approach may also lead to the issue of methods being classified into multiple areas, it adheres to the guidelines of DIN 18007 and the terminological framework established therein. According to this standard, the following demolition types are distinguished at the first level [102]:

- Mechanical methods (*dco:MechanicalMethod*);
- Thermal methods (*dco:ThermalMethod*);
- Hydrodynamic methods (*dco:HydrodynamicMethod*);
- Ablating methods (*dco:AblatingMethod*);
- Drilling methods (*dco:DrillingMethod*);
- Sawing methods (*dco:SawingMethod*);
- Splitting methods (*dco:SplittingMethod*);
- Blasting methods (*dco:BlastingMethod*).

Within the mechanical methods, for instance, techniques such as pulling and striking are described. The striking method can be further divided into elements like a sledgehammer or wrecking ball [102]. Consequently, a second hierarchical level can be established, to which additional instances can be assigned. The elements of this second level are also represented in accordance with the structure of DIN 18007. Using the object property *dco:willBeDeconstructedBy*, a building component to be deconstructed (e.g., an IFC object) can be semantically linked to the deconstruction methods.

The selection of a deconstruction method first requires the establishment of one or more criteria for the area of application. These criteria are represented as data properties, in alignment with DIN 18007, and are subdivided under the data property *dco:functionalApplicationArea*, *dco:materialApplicationArea*, and *dco:specificApplicationArea* (Domain: *dco:deconstructionMethod*). The material application area, for example, initially assigns a specific building component to a process. Typical representatives of this property include masonry or unreinforced concrete. In contrast, the functional application area characterizes the sub-goals associated with a method.

Another criterion that can be used to determine a method and is also anchored in DIN 18007 is the emission behavior. Emissions can vary in form and intensity [102]. Air pollutants in the form of dust and noise emissions can cause health impairments in humans. Furthermore, dust can contribute to environmental contamination and reduced visibility [102]. Vibrations and shocks can also lead to health issues and may impose additional stress on building components, potentially affecting their load-bearing capacity [91]. During deconstruction, additional emissions such as debris/material dispersal or water may occur [102]. These emissions are, therefore, incorporated as data properties linked to the class representing deconstruction methods (Domain: *dco:deconstructionMethod*). As sub-properties of *dco:levelOfExposure*, the data properties *dco:levelOfNoiseExposure*, *dco:levelOfVibrationExposure*, *dco:levelOfMaterialExposure*, and *dco:levelOfWaterExposure*.

4.4. Ontology Implementation

The implementation of the DCO ontology was carried out using the open-source software Protégé. This platform, which adheres to the standards of the World Wide Web Consortium (W3C), is built on Java, ensuring platform independence [108]. The DCO ontology, along with its integrated ontologies, consists of 327 classes, 89 object properties, 122 data properties, and approximately 4000 axioms. Figure 3 provides an excerpt from Protégé, where the "Classes" tab is selected. This tab allows for the creation and editing of ontology classes, and the class hierarchy is clearly visible. For instance, the selected class, *dco:CollapseBlasting*, is a subclass of *dco:BlastingMethod*, which is, in turn, a subclass of *dco:DeconstructionMethod*.

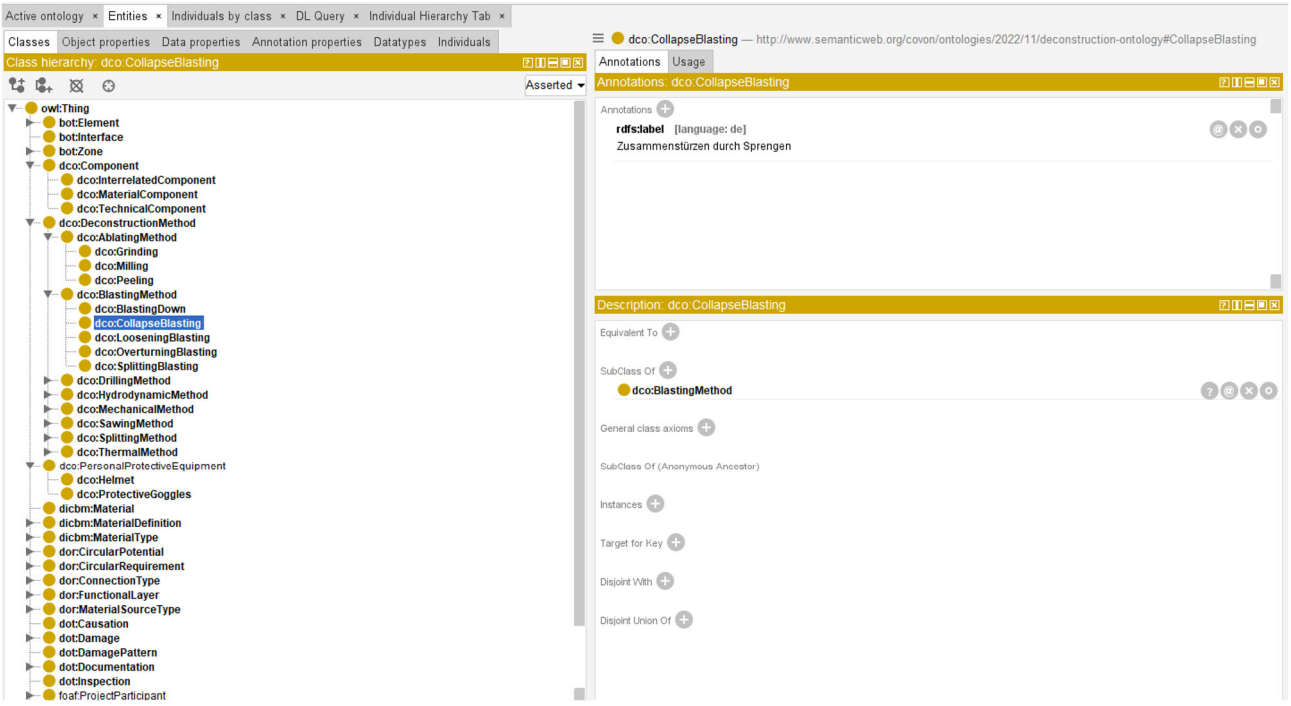


Figure 3. Excerpt of Ontology Editor Protégé.

5. Validation with Connection to a BIM Model

5.1. Integration of a BIM Model into a Graph Database Using Ontology

The following analysis focuses on the deconstruction of a single-family home. An exemplary overview of the subject is shown in Figure 4. Based on the premise that the residential building was constructed in 1955 and shows no signs of collapse risk, the subsequent analysis will consider its deconstruction. The structure primarily consists of

masonry and reinforced concrete. Additionally, there are no other structures present on the property. In the surrounding neighborhood, there are two other buildings of the same use type, which do not directly adjoin the deconstruction object. During the deconstruction process, emissions in the form of dust are expected, necessitating the recommendation for the enclosure of the building. The load-bearing components of the building include the exterior walls, the foundation, the floor slabs, and the roof structure. Only the walls located within the building envelope can be characterized as non-load-bearing. Regarding the construction period, it is noted that the exterior walls have a single-layer construction without thermal insulation. In the modeling process, attributes of the building components are assigned, also considering the fire resistance class. This information facilitates the assignment of further properties, such as potential contamination. It also enables further subdivision of materials, for instance, into inorganic and organic substances. The building model exported in the open exchange format IFC serves as the basis for the use and, thus, the validation of the DCO ontology in the next step. In this context, the approach of graph databases using Linked Building Data (LBD) is chosen, which is also applied in various research approaches [109]. The application of LBD allows the linking of BIM models with other datasets and the semantic knowledge representation through ontologies, making it suitable for the intended validation of the DCO ontology [109]. Graph databases are described as "systems specifically designed for managing graph-like data following the basic principles of database systems, i.e., persistent data storage, physical/logical data independence, data integrity, and consistency" [110]. The Stardog graph database is used in this research project [111].

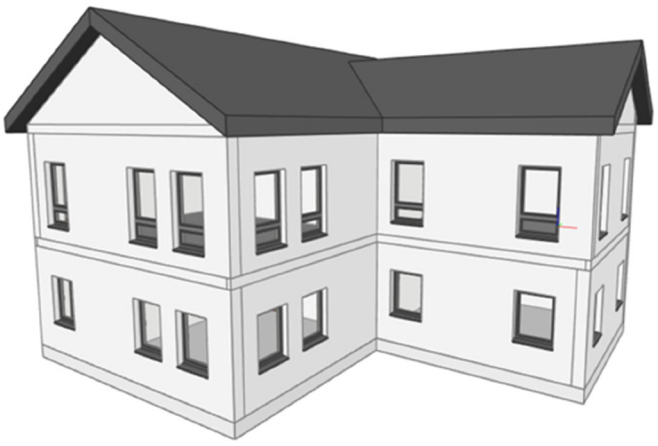


Figure 4. BIM model demonstrator.

The use of the graph database approach requires that all data used within the database be uploaded in an RDF-compliant format. Therefore, a conversion of the previously modeled IFC model into the Turtle (TTL) format is necessary, creating a network of knowledge graphs. TTL can be described as a concrete syntax of RDF [112]. The conversion is performed using the IFCtoLBD converter. This tool is designed to transform IFC building models into LBD, which is represented as RDF graphs. It uses the W3C Linked Building Data ontologies such as BOT, PRODUCT, and PROPS to structure the data. The converter extracts relevant information from IFC files and organizes it into modular RDF graphs that can be easily queried and analyzed. One of its key benefits is the generation of compact and well-structured graph representations that improve processing efficiency, especially for large BIM models. The open-source converter is implemented in Java using the Jena framework and supports all major IFC schema versions, from IFC2x3 TC1 to IFC4 Add2 [69].

Depending on the settings made, the conversion can be performed so that each IFC object is transferred into an instance. The resulting instance retains the modeled object attributes, such as the fire resistance class. Furthermore, the instances are directly assigned to the corresponding building classes using the W3C ontologies [69].

An exemplary generated instance of a BIM object "wall" is referenced with the identifier *inst:wall_a47a1543-0c11-465e-9ad3-833f396dab0d*. This instance is automatically assigned to the classes of the BOT and BEO ontology, establishing, for example, the connection *inst:wall_a47a1543-0c11-465e-9ad3-833f396dab0d* as "*rdf:type bot:Element*". Additionally, the material attributes are described by the ontologies used by the IFCtoLBD converter, such as *props:aussenmaterial_simple*. The load-bearing capacity of a component can also be defined through the mentioned ontologies, specifically through *props:tragwerksverwendung_simple*. Thus, the converted component retains the previously modeled properties and also receives further semantic information, such as the assignment to ontology classes.

To represent the domain of deconstruction processes and CE within the graph database, real data, such as the "gripper" as a deconstruction method, are directly created as instances in TTL format and uploaded to the database. This step serves to simplify the process and thus provides an alternative to conversion.

The DCO is specified for the demolition of buildings. The demonstrator is merely an example of an apartment building. The approach can also be transferred to other building types, e.g., office or industrial buildings. This is due to the fact that the DCO is based on a standardized topology and structure for the representation of buildings based on the ontologies shown in Table 1 (e.g., BOT). In addition, the DCO is based on established standards, guidelines, and guidelines for deconstruction applicable to all building types (see Table 2). This ensures that the approach is scalable.

5.2. Ontology Validation with SPARQL and SHACL

5.2.1. SPARQL

The validation of the DCO ontology is conducted using the graph-based query language SPARQL, which is applied to databases within the RDF schema [113]. The objective is to verify the correct conversion of the IFC model, particularly focusing on attributes such as contaminated materials. Additionally, it aims to demonstrate access to external ontologies and illustrate the use of the DCO ontology.

Query 01—Structure of the Building:

The first query pertains to the building demonstrator and aims to verify the correct conversion of all components and their attributes. This query specifically retrieves information on walls within the building, including their sub-elements, such as doors and windows. Furthermore, the query retrieves details on the family, type, and area of the walls and their sub-elements. It also differentiates between "load-bearing" and "non-load-bearing" walls. The results are critical for the planning of a deconstruction project, as they provide essential information about each object and the relationships between walls and doors/windows. The outcomes of this query are illustrated in Figure 5.

Query 02—Contaminated Materials:

Evaluating the secondary contamination of building components due to pollutant release requires knowledge of the types of pollutants involved. Secondary contamination in a building occurs during the dismantling phase when harmful toxic substances spread from the building material into the environment during the dismantling process or infect other installed materials [114–117]. This query retrieves all components containing pollutants. To ensure accurate identification, the query includes the Globally Unique Identifier (GUID), the type of pollutant, and the affected material. The results, shown in Figure 6, indicate that the Polyvinyl Chloride (PVC) flooring of the ceiling is contaminated with plasticizers,

and asbestos cement is present in the roof. Special precautions must be taken for these components during planned deconstruction activities.

```
#Abfrage 1: Struktur des Bauwerks (Wände)
SELECT ?GUIDWand ?FamilieUndTypWand ?FlächeWand ?Tragend ?GUIDFenster ?FamilieUndTypSubElement ?FlächeSubElement
Where {
    ?InstanzWand rdf:type beo:Wall.
    ?InstanzWand props:familieUndTyp_simple ?FamilieUndTypWand.
    ?InstanzWand bot:hasSubElement ?InstanzSubElement;
                 props:flXEche_simple  ?FlächeWand;
                 props:globalIdIfcRoot_attribute_simple ?GUIDWand;
                 props:tragwerksverwendung_simple ?Tragend.
    ?InstanzSubElement props:familieUndTyp_simple ?FamilieUndTypSubElement;
                       props:globalIdIfcRoot_attribute_simple ?GUIDFenster;
                       props:flXEche_simple  ?FlächeSubElement.

}
```

Run to File | Text | Charts | 18 Results, 623 ms

GUIDWand	FamilieUndTypWand	FlächeWand	Tragend	GUIDFenster	FamilieUndTypSubEl...	FlächeSubElement
"08nglwExf4WwQJEn$X...	"Basiswand: KS 175 + ...	19	"Tragend"	"2y$j2dYQX7xufzBJhrM...	"FE 1 tlg - DrehKipp: 11...	1
"08nglwExf4WwQJEn$X...	"Basiswand: KS 175 + ...	27	"Tragend"	"2y$j2dYQX7xufzBJhrM...	"FE 1 tlg - DrehKipp: 11...	1
"3JxuuJEPH6SRLSQX20...	"Basiswand: KS 175 + ...	14	"Tragend"	"3JxuuJEPH6SRLSQX20...	"FE 2 tlg - waagrecht ge...	2
"08nglwExf4WwQJEn$X...	"Basiswand: KS 175 + ...	27	"Tragend"	"2y$j2dYQX7xufzBJhrM...	"FE 1 tlg - DrehKipp: 11...	1
"08nglwExf4WwQJEn$X...	"Basiswand: KS 175 + ...	27	"Tragend"	"3JxuuJEPH6SRLSQX20...	"FE 1 tlg - DrehKipp: 11...	1
"3JxuuJEPH6SRLSQX20...	"Basiswand: KS 175 + ...	35	"Tragend"	"3JxuuJEPH6SRLSQX20...	"FE 2 tlg - waagrecht ge...	2
"3JxuuJEPH6SRLSQX20...	"Basiswand: KS 175 + ...	35	"Tragend"	"3JxuuJEPH6SRLSQX20...	"FE 2 tlg - waagrecht ge...	2
"3JxuuJEPH6SRLSQX20...	"Basiswand: KS 175 + ...	28	"Tragend"	"3JxuuJEPH6SRLSQX20...	"FE 2 tlg - waagrecht ge...	2
"2aUXL33156NfhJWpyv...	"Basiswand: KS 175 + ...	36	"Tragend"	"2y$j2dYQX7xufzBJhrM...	"FE 1 tlg - DrehKipp: 11...	1
"3JxuuJEPH6SRLSQX20...	"Basiswand: KS 175 + ...	17	"Tragend"	"3JxuuJEPH6SRLSQX20...	"FE 2 tlg - waagrecht ge...	2
"08nglwExf4WwQJEn$X...	"Basiswand: KS 175 + ...	19	"Tragend"	"2y$j2dYQX7xufzBJhrM...	"FE 1 tlg - DrehKipp: 11...	1
"2aUXL33156NfhJWpyv...	"Basiswand: KS 175 + ...	14	"Tragend"	"2y$j2dYQX7xufzBJhrM...	"FE 1 tlg - DrehKipp: 11...	1
"2aUXL33156NfhJWpyv...	"Basiswand: KS 175 + ...	14	"Tragend"	"2y$j2dYQX7xufzBJhrM...	"FE 1 tlg - DrehKipp: 11...	1
"3JxuuJEPH6SRLSQX20...	"Basiswand: KS 175 + ...	28	"Tragend"	"3JxuuJEPH6SRLSQX20...	"FE 2 tlg - waagrecht ge...	2
"2aUXL33156NfhJWpyv...	"Basiswand: GK 125"	13	"Nichttragend"	"2y$j2dYQX7xufzBJhrM...	"TU DF 1 - Rahmenstoc...	4
"3JxuuJEPH6SRLSQX20...	"Basiswand: KS 175 + ...	28	"Tragend"	"3JxuuJEPH6SRLSQX20...	"FE 2 tlg - waagrecht ge...	2

Figure 5. Result of Query 01.

```
#Abfrage 2: Kontaminierte Materialien
Select ?GUID ?Schadstoff ?TragendesMaterial ?Innenmaterial ?Außenmaterial ?Dämmschicht
Where {
    ?contaminatedMaterial props:schadstoff_simple ?Schadstoff;
                          props:globalIdIfcRoot_attribute_simple ?GUID;
                          props:tragendesMaterial_simple ?TragendesMaterial;
                          props:innenmaterial_simple ?Innenmaterial;
                          props:aussenmaterial_simple ?Außenmaterial;
                          props:daemmschicht_simple ?Dämmschicht.

}
```

Run to File | Text | Charts | 3 Results, 515 ms

GUID	Schadstoff	TragendesMaterial	Innenmaterial	Außenmaterial	Dämmschicht
"2y$j2dYQX7xufzBJhrME9Q"	"Weichmacher aus Tris(2-chl...	"Stahlbeton"	"Zementestrich"	"PVC Bodenbelag mit Weich...	"Trittschalldaemmung aus M...
"3JxuuJEPH6SRLSQX20SGQZ"	"Weichmacher aus Tris(2-chl...	"Stahlbeton"	"Zementestrich"	"PVC Bodenbelag mit Weich...	"Trittschalldaemmung aus M...
"3JxuuJEPH6SRLSQX20SUkr"	"Asbest"	"Fichtenholz"	"Dampfbremse aus Polyethyl...	"Asbestzement"	"Waermedaemmung aus Ste...

Figure 6. Result of Query 02.

Query 03—Reuse:

To ensure the sustainable use of deconstructed components, a circular potential analysis is conducted. This involves accessing the DOR ontology to identify components with their respective circular potential and the necessary requirements to realize this potential. Such requirements may include material quality assessments or logistical measures like non-destructive transportation (see Figure 7). For the results, the circular potential is presented along with the GUID for precise identification. For instance, a calcium silicate wall might be reusable, or a window could be recyclable.

```
#Abfrage 3: Wiederverwendung
SELECT ?GUID ?CircularPotential ?Requirement ?ClassOfRequierement
WHERE{
    ?x dor:hasCircularPotential ?CircularPotential;
       props:globalIdIfcRoot_attribute_simple ?GUID.
    OPTIONAL {
        ?y dor:hasRequirement ?Requirement;
           props:globalIdIfcRoot_attribute_simple ?GUID.
        ?Requirement rdf:type ?ClassOfRequierement
    }
}
```

Run to File | Text | Charts | 4 Results, 470 ms

GUID	CircularPotential	Requirement	ClassOfRequierement
"08nglwExf4WwQJEnXyMd"	inst:RepurposingPotential		
"3JxuuJEPH6SRLSQX20SJI0"	inst:ReusePotential		
"2y$j2dYQX7xufzBJhrMm8W"	inst:RecyclingPotential	inst:InspectionOfMaterialQuality	dor:Inspection
"3JxuuJEPH6SRLSQX20SUkr"	inst:DiscardPotential	inst:DamageFreeTransportOption	dor:Logistics

Figure 7. Result of Query 03.

Query 04—Deconstruction Methods:

The fourth query focuses on the deconstruction methods domain of the DCO ontology (see Figure 8). It identifies the planned deconstruction methods for various components. Additionally, it queries the material application and functional application to enable a detailed specification of the intended use. The results shown in Figure 8 consider the scenario of dismantling a masonry wall using a grabber. Moreover, the window is to be dismantled and completely removed.

```
#Abfrage 4: Rückbauverfahren
Select ?GUID ?DeconstructionMethod ?materialApplicationArea ?functionalApplicationArea
Where {
    ?x props:globalIdIfcRoot_attribute_simple ?GUID.
    ?x dco:willBeDeconstructedBy ?DeconstructionMethod.

    Optional {
    ?DeconstructionMethod dco:materialApplicationArea ?materialApplicationArea
    }

     Optional {
    ?DeconstructionMethod dco:funcionalApplicationArea ?functionalApplicationArea
    }
}
```

Run to File | Text | Charts | 2 Results, 1959 ms

GUID	DeconstructionMethod	materialApplicationArea	functionalApplicationArea
"08nglwExf4WwQJEnXyMd"	inst:gripper	"masonry"	
"3JxuuJEPH6SRLSQX20SJI0"	inst:dismantling		"removing"

Figure 8. Result of Query 04.

5.2.2. SHACL

Shape 01—Load-Bearing Elements:

In practice, distinguishing between load-bearing and non-load-bearing components is crucial for safe deconstruction [106]. Load-bearing components significantly contribute to the structural stability of a building. Improper handling during deconstruction can pose safety risks. To minimize such risks, a SHACL (see Table 3) rule is established to issue a warning if the *props:tragwerksverwendung_simple* attribute is marked as "tragend" (en: load-bearing). This allows for early identification and proper handling of load-bearing components during the deconstruction process.

Table 3. SHACL Syntax for the Shape 01—Load-Bearing Elements.

```
#BearingElements
dco:ConstructionShape
    a sh:NodeShape;
    sh:targetClass bot:Element;
    sh:property [
        sh:path props:tragwerksverwendung_simple;
        sh:datatype xsd:string;
        sh:not [sh:in ("Tragend")];
        sh:message "caution deconstruction will have an impact on stability"@en;
        sh:severity sh:Warning
    ].
```

Shape 02—Fire Rating:

The fire behavior of building components is another critical aspect during deconstruction, as certain procedures may involve heat generation or sparks, increasing the risk of fire [94]. A SHACL rule is introduced to ensure that components with less than 30 min of fire resistance are identified (see Table 4). In such cases, an informational message is generated, indicating the need for additional protective measures.

Table 4. SHACL Syntax for the Shape 02—Fire Rating.

```
#FireRating
dco:FireRatingShape
    a sh:NodeShape;
    sh:targetClass bot:Element;
    sh:property [
        sh:path props:fireRating_simple;
        sh:datatype xsd:string;
        sh:not [sh:in ("F0")];
        sh:severity sh:Info;
        sh:message "attention the material has no fire resistance"@en;
    ].
```

Shape 03—Contamination:

Similar to the second SPARQL query, the SHACL query 03 (see Table 5) rule checks components for potential contamination. The rule aims to identify all elements marked with the property *props:schadstoff_simple* as contaminated. For these components, a warning is issued to highlight the need for special precautions during handling and disposal.

Table 5. SHACL Syntax for the Shape 03—Contamination.

```
#Contamination
dco:ContaminationShape
    a sh:NodeShape;
    sh:targetClass bot:Element;
    sh:property [
        sh:path props:schadstoff_simple;
        sh:maxCount 0;
        sh:severity sh:Warning;
        sh:message "caution the material is contaminated"@en;
    ].
```

6. Discussion

The article presents the development and validation of an ontology for the deconstruction of buildings and the possibility of reusing components for new construction projects. First, a literature analysis is conducted on related works in the areas of BIM and building deconstruction, as well as existing ontologies in building construction and sustainability. Subsequently, the ontology is developed according to the Poveda-Villalón approach. Firstly, the ontology requirements were specified. In addition, existing ontologies were identified for integration into the DCO. The specification and the identified existing ontology were then used as the basis for the actual development of the DCO and its implementation within an ontology editor. In addition, the ontology was published, and an approach for continuous maintenance was presented. For validation, the DCO was linked to a digital 3D model, and four questions were queried using SPARQL and three rules using SHACL.

By successfully developing and validating the DCO, a contribution was made to the formalization and representation of existing knowledge in the life cycle phase of building demolition. As a result, a standardized understanding can be established for all participants in the deconstruction phase as well as the preliminary operating phase and following new project development. By linking the DCO with a 3D model and the successfully completed queries, the usability and performance of the ontology are demonstrated. This supports the further use of digital 3D models according to the BIM method as well as the interoperability between heterogeneous as-built data and systems in the phase cycle. At the same time, the ontology supports improved planning and implementation of demolition measures. In addition, the DCO promotes the reusability of components or the building fabric in the sense of the CE by making it possible to precisely analyze and depict existing potentials and possible terminated danger zones during the deconstruction process. Moreover, the correct application of the approach and the associated optimal planning and execution of deconstruction measures also contribute to the reduction of CO_2 emissions (ecological sustainability). As a result, materials are not simply left unused or incinerated during deconstruction but are optimally deconstructed and sorted so that they are as clean and reusable as possible. This guarantees easier reintroduction into the cycle of new buildings. Furthermore, social and economic sustainability are also improved. Social sustainability is achieved through improved protection of personnel, and economic sustainability is achieved through optimal cooperation as well as process and resource planning.

A further added value of the DCO is the use of existing ontologies (see Table 1) and the extension with the specific domain knowledge of the planning and execution of demolition measures of existing buildings on standards, guidelines, and regulations, as well as leaflets (see Table 2). There are no redundancies to existing ontologies in the field of sustainability and circular economy, but rather a substitution and expansion of the existing knowledge base. The articles of Akbarieh et al. should be considered in particular [16,66,86]. In this context, there are no redundancies in the knowledge structures of the two ontologies but rather a supplementation and utilization of existing synergies. During the development of the DCO, all existing ontologies were analyzed, and only the knowledge missing in the specific area of planning and execution of building demolition processes was formalized and integrated into the structure of the DCO.

Although the usability of the DCO has been successfully developed, there are also limitations in the research approach. To date, the ontology has only been based on explicit knowledge from specialized literature, such as standards, technical regulations, and specifications. An integration of existing implicit knowledge from industry practice by experts is not yet provided. The integration of further ontologies of the previous and subsequent life cycle phases of the building is also required. This can limit the acceptance of end users in practice and the full utilization of the DCO. Furthermore, the validation was conducted on

a 3D model as a demonstrator. As a result, the demonstrator, including the integrated data sets, was accurate and adequate for testing. In this context, real data sets with a certain imperfection are missing so that the linking and the model application are functional even under different degrees of quality. The ontological approach using the RDF data schema has significant advantages for data usage, such as machine interpretability, linking, and interoperability. Nevertheless, the simple transfer to industry is not feasible. This is a further limitation. At the current state of the art, DCO cannot be integrated into established project management systems and other software applications without further adaptation research and development steps. Furthermore, existing material databases need to be transformed from a relational database schema into a graphical database schema.

Further research is required to overcome the limitations and further improve the ontology. For example, the missing implicit knowledge of the ontology can be acquired through qualitative studies, such as expert interviews, and integrated into the ontology in a further round of optimization. In addition, further ontologies of different scenarios along the life cycle need to be analyzed and integrated to ensure interoperable, transparent, and multi-stage data management of the building. Furthermore, a more comprehensive validation should be carried out in the course of a sensitivity analysis with real data sets and different building types as 3D models. The use of different types and qualities of data sets is the only way to determine the performance and existing limitations of the ontology. In addition, existing systems, such as the IFC2LBD-Converter, must be further developed. There is also a need to establish further interfaces and tests that integrate and link the ontological approach with practical application systems and database systems.

7. Conclusions

This paper presents the development of the DCO, an ontology for the deconstruction of buildings and the reuse of individual components and materials. The ontology is stepwise conceptualized; existing ontologies are integrated and validated using a demonstrator. In conclusion, the approach is discussed, and limitations and future research activities are indicated. The approach represents a new possibility of knowledge formalization and representation for the deconstruction phase of buildings. This represents a further possibility of life cycle-oriented data utilization by linking and using open data exchange formats without using the limited IFC data format. It increases interoperability and offers the possibility of gaining further insights into reuse in the context of the CE in combination with additional data relating to materials and sustainability. This is a further step towards the circular management of buildings. This contributes to the better integration of sustainability aspects in the construction industry, the avoidance of material waste, and the conservation of resources. In addition, all these aspects contribute to increasing the degree of digitalization and automation. In future research work, the ontology will be further specified and optimized in several iterative tests. Moreover, it will be linked to other life cycle phases of buildings and further detailed. This implies the linking to the ontologies of planning, construction, and operation of buildings, as well as the development of new buildings. This allows data records to be linked across the individual phases. As a result, algorithms can interpret data across the life cycle and identify previously hidden relationships between different data sets. This provides the basis for holistic knowledge management in the future and simplifies the use of AI systems.

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